

Implement additively manufactured crack-resistant material inserts at rail joints to prevent hydrogen degradation and fatigue fracture

Executive Technology Brief

Domain: mechanical_structural

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'rail steel fatigue fracture' (broken rails/joints, FRA cause family T2): 3 of 25 FRA Form 54 accidents in the retrieved window are broken-rail/joint-caused, derailing 17 cars total. Causes observed: 'Broken Rail - Base [Desc.Changed-Refer to Current/Prev.Guides]'; 'Mismatched rail-head contour'. LIMITS STAMPED INLINE: carrier-reported regulatory reports; cause codes are classifications of the initial report, NOT proven root causes; reportable-above-threshold events only (Art. XXI.5 analog).

Mechanism: Hydrogen absorption in pearlite steel leads to reduced fatigue life and decreased fracture toughness

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Implement additively manufactured crack-resistant material inserts at rail joints to prevent hydrogen degradation and fatigue fracture

Governing Model: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: CONDITIONAL; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value – geometry factors must be applied from the actual design. Engaged variables: NOT ESTABLISHED.

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: CONDITIONAL; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation – use their measured plateau relations instead. Engaged variables: NOT ESTABLISHED.

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.